

■ AI-Powered College Timetable Generator

Enterprise-grade university timetable automation powered by Google OR-Tools CP-SAT solver.

Fully responsive · Cross-platform · PDF & Excel export · Dual semester · Staff workload tracking

■ Three Scheduling Modes — All Verified

Mode	Description	Status
Auto Only	Add data → click Generate. Solver handles everything.	✓ Works
Auto + Fixed Slots	Pre-lock some slots → solver fills the rest.	✓ Works
Assignments + Fixed Slots	Manually assign staff hours → solver respects all.	✓ Works

■ Key Features

Staff Workload Assignments: Manually assign teachers to subjects with weekly hours. Split subjects across multiple teachers. Live red-border validation + toast errors for constraint violations.

Lab Split Scheduling: 2h→1×2 block, 3h→1×3 block, 4h→2+2 days, 5h→2+3 days, 6h→3+3 days. At most 1 lab per class per day.

Fixed Slot Pinning: Lock any subject to a specific day & period. Supports bulk pinning for all sections of a department/year. Solver always respects these first.

Dual Semester Syllabus: Track Odd (1,3,5) and Even (2,4,6) semester syllabi per class. 1-click switch between semesters.

Language Capping: Tamil/English subjects automatically capped at max 1 hour per day.

PDF & Excel Export: Class-wise sorted by year (1st→2nd→3rd→PG). Staff-wise sorted by Staff ID. Big name heading + hours format: Name (Theory + Lab = Total).

Diagnose Tool: Pre-flight check: detects staff capacity deficits, missing assignments, fixed slot collisions before generating.

Backup & Restore: Full JSON backup & 1-click restore. Excel import template for bulk data entry.

Bulk Syllabus Copy: Copy one class syllabus to all sections of same department & year in 1 click. UG/PG strictly isolated.

Cross-Platform & Responsive: Works on Windows, macOS, Linux, Android (Termux), iOS (a-Shell). Fully responsive UI. Share via Wi-Fi on same network.

■ University Period Timings

Period	Time
Period 1	09:20 – 10:10
Period 2	10:10 – 11:00

Period 3	11:00 – 11:50
Lunch Break	11:50 – 12:30
Period 4	12:30 – 01:10
Period 5	01:10 – 02:00

■ Project Files

File	Purpose
app.py	Flask backend — all API routes and data management
solver.py	CP-SAT timetable solver engine (Google OR-Tools)
exporter.py	PDF (ReportLab) & Excel (openpyxl) export functions
templates/index.html	Full single-page frontend UI (HTML + CSS + JS)
data.json	Live database — auto-saved on every change
requirements.txt	Python package dependencies
README.md	Setup and usage guide

■ Quick Start

Install:

```
pip install -r requirements.txt
```

Run (macOS/Linux):

```
python3 app.py
```

Run (Windows):

```
python app.py
```

Open **http://127.0.0.1:5000** (or whichever port is shown) in any browser.